

Landscape Overview

Task 3 of the MRiXFields2026 challenge calls for a **single unified model** that can translate brain MRI scans between *any* pair of field strengths (0.1T–7T) and modalities. Traditional GAN-based pipelines (e.g. CycleGAN) have been used as baselines [29†L87-L90], but recent advances emphasize **diffusion-based and foundation models** that handle multiple inputs/outputs flexibly. In practice, top-performing approaches adopt large-scale generative models (diffusion or hybrid) with explicit structural or semantic guidance. For example, latent-space diffusion models (e.g. HCLD [40†L63-L71]) explicitly disentangle content/style to preserve anatomy, while anatomy-guided diffusion (e.g. BrainMRDiff [55†L50-L59]) and segmentation-guided models (e.g. TAM [44†L185-L193]) condition generation on masks to ensure anatomical fidelity. Other recent work reframes multi-modal MRI synthesis as a single *inpainting* task (Any2all [48†L231-L239]) or uses contrastive/CLIP guidance to align outputs semantically (e.g. a CLIP-based foundation model [46†L20-L28]). In summary, the current frontier blends **latent diffusion**, **adversarial diffusion**, **anatomy conditioning**, and **foundation-learning** to achieve robust, multi-domain MRI translation with high fidelity.

Approach Categories

- **GAN-based Multi-domain Translation:** Early methods like **CycleGAN** (paired with Cycle-consistency) and **StarGAN v2** handle multi-site translation with single-model architectures. These use adversarial losses to match styles across fields, but often struggle with anatomical detail at severe domain shifts [29†L96-L99]. *Baselines:* MRiXFields’ own baseline uses StarGAN v2 for Task 3 [29†L96-L99]. Multi-modal GANs (mmGAN, mmResViT) and their extensions (ResViT [36†L31-L34]) are also relevant as competitive baselines for multi-domain MRI.
- **Latent-space Diffusion Models:** Methods like **HCLD** (Conditional Latent Diffusion) explicitly encode MRIs in a learned latent space and perform diffusion there [40†L63-L71]. HCLD separates content (normalized latent) and style (AdaIN conditioning), then learns a conditional latent diffusion (cLDM) to translate between styles. Similarly, **MISA-LDM** uses a 3D latent diffusion with a structure-preserving module (SPM) to ensure modality-invariant anatomical features [42†L47-L55]. These latent diffusion methods are powerful for 3D volumes (volume-level harmonization) and can preserve content across domains. *Strength:* faithful anatomy preservation and efficient 3D training [40†L63-L71] [42†L47-L55]. *Weakness:* still require costly sampling and training on large datasets; may oversmooth fine textures.
- **Cross-conditioned Diffusion Models:** **CDM** uses a novel two-step process: first learn a target-modality distribution via a lightweight diffusion network, then condition a U-Net on that distribution and the source image [51†L28-L37]. This “cross-conditioned” design offers faster sampling by decoupling heavy denoising from generation. Another variant is **UMH** (Unpaired MRI Harmonization) which does coarse latent diffusion for multi-site alignment, then a *CLIP-based fine-tuning* to refine style [32†L47-L56]. These methods emphasize **explicit conditioning** on desired style or content (image style embeddings, anatomical masks, etc.) to guide the diffusion. *Strength:* leverage semantic

or style guidance (e.g. CLIP) for better realism [32†L47-L56] . *Weakness*: multi-stage pipelines can be complex and may require pre-trained encoders (e.g. CLIP) which add overhead.

- **Anatomy/Segmentation-Guided Synthesis**: Models like **TAM** (“Translate Any Modality”) use segmentation networks to extract soft-tissue and bone masks, then feed them into a conditional diffusion model to anchor anatomy [44†L185-L193] . **BrainMRDiff** explicitly injects multiple anatomical masks (white matter, gray matter, ventricles, tumor) as controls and uses topology-preserving losses (persistence diagrams) to ensure structural consistency [55†L50-L59] . These approaches treat anatomy maps as conditional inputs (ControlNet style) to the generation process. *Strength*: Very high anatomical fidelity and clinical plausibility [44†L185-L193] [55†L50-L59] . *Weakness*: Relies on accurate segmentations; more complex models (e.g. topology loss) increase implementation difficulty.
- **Hybrid & Adversarial Diffusion**: **SynDiff** is a notable example that fuses GAN and diffusion ideas [35†L242-L251] . It uses large diffusion steps with adversarial projections and cycle consistency to enable unpaired training. Unlike pure GANs, SynDiff achieves markedly higher fidelity in MRI translation [35†L242-L251] [35†L253-L258] . Other hybrids (e.g. discrete diffusion with auxiliary GAN losses) fall in this category. *Strength*: Combines GAN sharpness with diffusion diversity; generally outperforms plain GANs [35†L242-L251] [35†L253-L258] . *Weakness*: Very heavy: running a diffusion model with GAN training is computationally expensive.
- **Unified/“Foundation” Models**: Inspired by large-scale models, some works train *single* models for “any-to-any” translation. The **Any2all** framework treats all modalities as a “virtual stack” and trains an **unconditional latent diffusion** to model that full stack [48†L231-L239] . At inference it performs multimodal inpainting (guided reverse sampling) to fill missing fields. Similarly, Pu et al. train a **single foundation model** with contrastive and CLIP-based learning to support any-to-all MRI synthesis [46†L20-L28] . These approaches emphasize *one model for everything*, potentially benefiting from massive data. *Strength*: Ultimate flexibility and consistency across modalities; may enable large pre-training. *Weakness*: Require huge datasets and compute; novel training pipelines (e.g. diffusion posterior sampling [48†L275-L283]) and careful guidance strategies.

Comparative Matrix

Model / Method	Approach	Strengths	Weaknesses	Compute Cost	Scalability
StarGAN v2 [29†L96-L99] (GAN)	Multi-domain GAN (single model)	Efficient training/ inference; supports many domains via style vectors. Baseline for Task 3 [29†L96-L99] .	Can <i>mode collapse</i> or produce inconsistencies across fields; less anatomical fidelity at extreme domain gaps.	Low-Medium (GAN-scale)	High (multi-domain)

Model / Method	Approach	Strengths	Weaknesses	Compute Cost	Scalability
CycleGAN [29†L96-L99] (GAN)	Unpaired cycle-GAN for pairwise fields	Simple unpaired training; proven on many datasets.	Limited to fixed input→output; needs many models for many fields. Hallucination of features possible.	Low	Poor (requires many models)
SynDiff [35†L242-L251] [35†L253-L258] (Hybrid)	Cycle-consistent Adversarial Diffusion	Superior image fidelity and diversity to pure GANs; supports unpaired data [35†L242-L251] [35†L253-L258] .	Very expensive training (diffusion sampling + GAN losses); complex training procedure.	Very High	Moderate (designed for MRI×CT, scalability unclear)
HCLD [40†L63-L71] (Diffusion)	Latent-space Conditional Diffusion	Preserves 3D anatomy by fusing source/target latents; volume-wise consistency [40†L63-L71] . Efficient two-stage training (autoencoder+diffusion).	Requires careful style/content loss design; still needs iterative sampling. Limited to T1 for now.	High (latent diffusion)	Moderate (designed for multi-site)
MISA-LDM [42†L47-L55] (Diffusion)	Latent Diffusion with Structural Prior	Many-to-many multi-contrast translation in one model; structural SPM retains topology [42†L47-L55] . Handles arbitrary missing inputs.	Increased model complexity (SPM, HFCM); training cost higher than GANs.	High (latent diffusion)	High (handles many combos)
CDM [51†L28-L37] (Diffusion)	Cross-conditioned Diffusion	Extremely efficient for conditional tasks: samples latent target distribution then guides U-Net. Improves quality with target distribution guidance [51†L28-L37] .	Two-phase design adds complexity; requires pretraining of representation model (MRM).	High (diffusion + distill)	High (any source→any target)

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BrainMRDiff 【55†L50-L59】 (Diffusion)	Anatomy-conditioned Diffusion	Topology-aware: explicitly enforces structural fidelity in tumors via persistent homology; multi-mask conditioning 【55†L50-L59】 . Very realistic anatomies.	Needs ground-truth masks for all structures; complex loss (topological) slows training/ inference.	High (diffusion + complex losses)	Low (specifically for tumor MRI)
TAM 【44†L185-L193】 (Diffusion)	Segmentation-guided Diffusion	Two-stage: UNet segmentation + diffusion ensures anatomy correctness and high image quality 【44†L185-L193】 . Demonstrated across many translation tasks (MRI↔CT) with quantitative gains.	Needs high-quality segmentations; two-model pipeline doubles training and inference cost.	Very High (3D UNet + diffusion)	High (multi-sequence inputs)
Any2all 【48†L231-L239】 (Diffusion)	Unified Latent Diffusion (Any→All)	One unconditional diffusion models <i>all</i> modalities simultaneously. Flexible inference (inpainting with known inputs) 【48†L231-L239】 . Unified training across tasks.	Requires massive data to model full modality stack; inference requires careful posterior sampling (e.g. DPS/MPS).	Very High (1 model, multivariate)	High (general tasks)
Any-to-All NPC 【46†L20-L28】 (Foundation)	Contrastive + CLIP-guided	End-to-end any→any model leveraging vision-language semantics. Strong generalization (trained on 40k+ images, robust to noise) 【46†L20-L28】 . Improves downstream tasks.	Very large-scale (data/ hardware); training required to align vision and language. Novel architecture complexity.	Extremely High (foundation-scale)	Very High (foundation model)

Model / Method	Approach	Strengths	Weaknesses	Compute Cost	Scalability
ResViT [36†L31-L34] (Hybrid)	CNN + Vision Transformer GAN	Captures both local detail (CNN) and global context (ViT); improves upon plain CNN generators.	Adds ViT overhead; still GAN-based with typical GAN issues (mode collapse, training instability).	High	Moderate (extends GANs)
mmGAN / mmResViT / M2DN (GAN/Diffusion hybrids)	Multi-modal GAN/Diffusion	Baseline architectures specialized for multi-contrast MRI. Useful points of comparison.	Generally inferior to newer diffusion methods.	Varies (medium)	Limited (static modality sets)

Expert Verdict

Most Promising: *Diffusion-based methods with explicit conditioning.* Our review suggests that hybrid/diffusion approaches (like SynDiff and TAM) and latent diffusion models (HCLD, MISA-LDM) consistently outperform vanilla GANs on image fidelity [35†L242-L251] [42†L47-L55]. In particular, **explicit anatomical or style conditioning** is crucial: TAM’s use of segmentation masks and BrainMRDiff’s topology loss yield highly accurate anatomies [44†L185-L193] [55†L50-L59]. Similarly, latent diffusion schemes (HCLD) avoid training multiple view-specific networks and preserve 3D structure better than slice-wise GANs [40†L63-L71].

Among **unified models**, Any2all and other foundation-style approaches are exciting because of their flexibility. Any2all’s single-model diffusion is conceptually elegant [48†L231-L239], and the CLIP-based foundation model demonstrates the benefit of large-scale contrastive/semantic training [46†L20-L28]. However, these “behemoth” solutions demand huge data and compute resources. In practice, a **two-stage strategy** like TAM (segmentation→diffusion) may be more tractable: it enforces anatomy (via a fast UNet) and then refines with diffusion, combining speed and quality [44†L185-L193].

Next Steps: Hybrid approaches that combine the best of these worlds appear most practical. For example, one could adopt a diffusion backbone (for high quality) but guide it with pretrained encoders (e.g. a segmentation or CLIP network) to reduce required training data. Also, leveraging **paired volunteer scans** (as MRixFields provides) to fine-tune these models could drastically improve anatomical consistency. Finally, thorough uncertainty quantification (e.g. comparing multiple diffusion samples) and rigorous generalization testing will be needed to ensure clinical robustness.

In summary, the current state-of-art favors *conditional diffusion models*, especially those incorporating anatomical priors or large-scale contrastive pretraining [51†L28-L37] [46†L20-L28]. These methods combine strong image quality with the flexibility required for “any-to-any” MRI synthesis, making them prime candidates for Task 3.

Sources: Recent literature and challenge resources have been cited above to substantiate all points.
